

TITLE: 110 GHz Cryogenic Coax Cables

Requesting Laboratory / Division / Group:

Communications Technology Laboratory
RF Technology Division
Superconducting Electronics Group (672.06)

General Statement of Need:

We are requesting items to increase the bandwidth of and existing dilution refrigerator wire to 110 GHz. This requires low thermal conductance (stainless steel or CuNi) thin-diameter coaxial wiring connectorized with 1 mm (or similar) rf connectors. Additionally, we request infrastructure to install this coax into an existing dilution refrigerator – i.e. a vacuum feedthrough and high thermal conductivity copper plates at each temperature stage. Suppliers will work with end-user to get exact dimensions and lengths.

Item Specifications and Quantities:

Item 1:	Cryogenic-Compatible Coax Cable	
Quantity:	10 MINIMUM (Various Lengths)	
Specifications:	1.1	Stainless-steel outer, copper or stainless-steel inner coax
	1.2	110 GHz bandwidth rated connectors (1mm preferred)
	1.3	Lengths decided for installation in dilution refrigerator in collaboration with end-user
Item 2:	Cryogenic-Compatible Coaxial Cable	
Quantity:	2 MINIMUM (Various Lengths)	
Specifications:	1.1	Superconducting Niobium-Titanium (NbTi) outer/inner coax
	1.2	110 GHz bandwidth rated connectors (1mm preferred)
	1.3	Lengths decided for installation in dilution refrigerator in collaboration with end-user
Item 3:	Hermetic Feedthrough and Non-Hermetic Feedthrough	
Quantity:	1 EACH	
Specifications:	3.1	One (1) KF25 or larger vacuum/hermetic feedthrough with 1mm connectors to allow connection of two 1 mm external cables on either side
Item 4:	Non-Hermetic Feedthrough	
Quantity:	4 EACH	
	4.1	Non-hermetic feedthroughs for 50 K, 4 K, still, and mixing chamber with two (2) 1 mm feedthroughs for installation of items 1 and 2.
	4.2	Main plate shall be Oxygen-Free High Conductivity (OFHC) Copper

[List any additional item(s) or service requirements using the same format.]

Procurement Support Information and Ancillary Services:

[Provide input for each section Below or Mark Not Applicable (N/A)]

Section	Requirements/Specifications
Planning Considerations ☐ N/A	☐ List any visits required to allow potential vendors to see site specific constraints. [Explain Requirements.] ☐ US citizen required on-site (foreign nationals require pre-registration 30 days prior to visit.) ☐ Vendor Representative On-Site more than three (3) days. ☒ Other Considerations: For quote purposes, a 20 inch length of coax can be used, but final lengths shall be coordinated with end-user
Shipping & Delivery ☐ N/A	Shipping Address: ☒ NIST-Boulder Mail Stop: 325 Broadway Boulder, CO 80305 ☐ NIST-Gaithersburg ATTN: Mail Stop: 100 Bureau Dr Gaithersburg MD 20899 ☐ [Alternate Shipping Address] Shipping Criteria: ☐ Partial Delivery Acceptable ☐ [Enter Shipping and Delivery information by item.] ☐ Direct Delivery to OU Building Required [Explain Need for Requirement] Building /Room Number: [Enter Building/Room Number] ☐ Building has Loading Dock Containerization Preference: [Enter Container Type]

Other Requirements:

[List Any Other Requirement]

Delivery Date Criteria: 90 days ARO

☒

☐ Delivery Shall Be Made No Earlier Than:

Click or tap to enter a date.

☐ Scheduled Deliveries:

Item Number	Due Date

☐ Other:

[List Other Shipping, Delivery or Special Requirements.]

Electronic Media

☒ N/A

☐ Software

☐ Electronic Manuals

☐ License Key

☐ Other:

[List Any Other Electronic Media Requirements]

☐ US Government Email Address for Software Delivery:

[Enter Address]

[Enter Electronic Media Requirements by Item.]

Installation

☒ N/A

☐ Vendor Technician on Site

☐ Rigging

☐ Uncrating / Unpackaging

☐ Removal of Packaging Material

☐ Equipment Set Up

☐ Start-Up Services

☐ Turnkey Installation

☐ Other:

[Enter Details]

[Enter Installation Requirements by Item]

NOTE: SME shall be the coordination POC.

Facility / Utility Considerations

☒ N/A

☐ OFPM Work Order Numbers

[List Work Order Numbers]

☐ Power Available

Voltage:

[Enter voltage] (V)

Maximum Current:

[Enter max current draw] (A)

Phase:

[Choose Phase]

Frequency:

[Typically 60Hz] (Hz)

NEMA Plug Type:

[Enter Plug Type] (i.e.: NEMA L5-20P)

Environment:

[Choose Environment]

Other Power Considerations:

[List Any Other Power Considerations.]

☐ Utility Connections:

[List Relevant Available Utilities, Pressure(s) and Flowrate(s).]

☐ Other Considerations:

[List Other Considerations to Ensure an Accurate Quote.]

[List Site Specific Utility Considerations by Item]

Inspection and Acceptance

☐ N/A

☒ On-Site

☒ Acceptance Not Expected to Exceed 7 Business Days

☐ Acceptance Expected to Exceed 7 Business Days:

[Duration]

☐ Inspection and Acceptance Plan:

[Explanation]

☒ Other:

[Inspection Requirements Per Item (Who, When, What, Where)]

Training

☒ N/A

☐ Number of Trainees:

[Enter Number of Trainees]

☐ On-Site

☐ Off-Site

Location:

[Enter Location(s)]

Is Trainee Travel Required:

[Choose Answer]

Scope of Training

☐ Operation

☐ Maintenance

☐ Troubleshooting

☐ Safety

☐ Other

[Enter Other Scope of Training Requirements]

☐ Training Materials to be Provided by Government:

Training Requirements (Who, When, What, Where)

	[Describe Materials] ☐ Training Materials to be Provided by Vendor: [Describe Materials] ☐ Other [Enter Other Training Requirements] NOTE: SME shall be coordination POC.
Warranty ☐ N/A	Base Manufacturer Warranty is acceptable Warranty Length: [i.e. 3 - Years] What is included? ☐ Labor ☐ Parts ☐ Travel ☐ Cost and Liability for Returns ☐ Other [Enter Other Requirements] ☐ 1 Option Period ☐ 2 Option Periods ☐ 3 Option Periods ☐ 4 Option Periods (Maximum)
Calibration ☒ N/A	[Describe Calibration Requirements Per Item] Calibration Plan Length: [i.e. 3 - Years] What is Included? ☐ Labor ☐ Spare Parts ☐ Consumables ☐ Travel ☐ Cost and Liability for Shipment ☐ Software Updates ☐ Manuals ☐ Other [Enter Other Requirements] ☐ Requirements: [Enter Requirements] (i.e.: On-Site Within 72 hours, etc.) ☐ 1 Option Period Length: [i.e. 1 – Year / 6 - Months] ☐ 2 Option Periods ☐ 3 Option Periods ☐ 4 Option Periods (Maximum)
Maintenance ☒ N/A	[Describe Maintenance Requirements Per Item] Maintenance Plan Length: [i.e. 3 - Years] What is Included? ☐ Labor ☐ Spare Parts ☐ Consumables ☐ Travel ☐ Cost and Liability for Shipment ☐ Software Updates ☐ Manuals ☐ Other [Enter Other Requirements] ☐ Requirements: [Enter Requirements] (i.e.: On-Site Within 72 hours, etc.) ☐ 1 Option Period Length: [i.e. 1 – Year / 6 - Months] ☐ 2 Option Periods ☐ 3 Option Periods ☐ 4 Option Periods (Maximum)
Government Furnished Material / Equipment ☒ N/A	List Any Government Furnished Material / Equipment Per Item] ☐ Samples for Testing ☐ Equipment which will leave Government Site Description: [Enter Description] Serial Number: [Enter Serial Number] NIST Property Number: [Enter NIST Property Number] ☐ Other: [Enter Other Details]
Travel ☒ N/A	[Describe any Travel Requirements Per Item] ☐ Explain: [Enter Justification / Requirement] ☐ Other: [Enter Other Requirements] NOTE TO VENDOR: To the maximum extent possible, include travel requirement estimates as part of pertinent sections above (Installation, Maintenance, Repair, etc.). Otherwise, Travel will be a separate line item, reimbursable per the Federal Travel Regulation.